INCH-POUND

MIL-A-47312A (MI)
16 December 1992
SUPERSEDING
MIL-A-47312 (MI)
30 September 1987

MILITARY SPECIFICATION

ACTUATOR, HYDRAULIC

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the requirements of one type of hydraulic actuator controlled by a four-way electrohydraulic flow control servo valve.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1420

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-P-45213	-	Preservation and Packing of Rocket and Missile Systems Equipment for Shipment
MIL-V-47308	-	Valve, Servo, Hydraulic Flow Control

STANDARDS

MILITARY

MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-970	-	Standards and Specifications, Order of Precedence for the Selection of
MS3116	-	Connector, Plug, Electric, Series 1, Solder Type, Straight, Bayonet Coupling
MS33649	-	Bosses, Fluid Connection-Internal Straight Thread

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this

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document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Materials. Materials and finishes to be used in construction of the actuator shall be selected in accordance with the order of precedence specified in MIL-STD-970.

3.3 Design and construction. The design and construction of the actuator shall be in accordance with this specification and figure 1 (see 4.6.1).

3.3.1 Weight. Weight of the actuator shall be not greater than 4.0 pounds (see 4.6.1).

3.3.2 Fluid media. The actuator shall be designed to operate with hydraulic fluid conforming to MIL-H-5606 (see 4.6.1).

3.4 Performance.

3.4.1 Stroke. The actuator stroke shall be as shown in figure 1 (see 4.6.2).

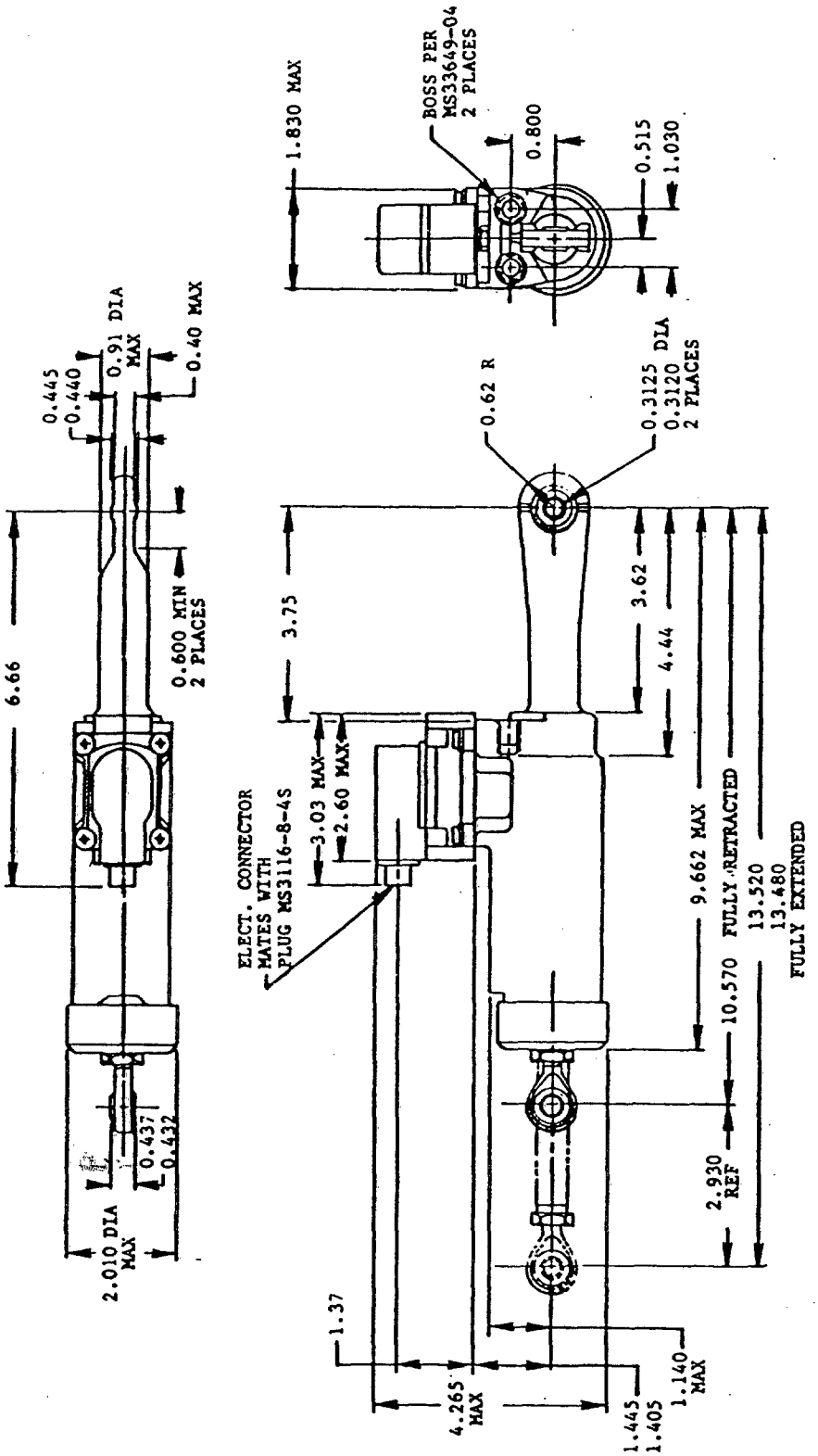
3.4.2 Displacement polarity. A rated command to the servo valve causing flow out of valve port 1 shall extend the actuator (see 4.6.2).

3.4.3 Velocity. The actuator shall have a minimum zero load velocity of 4.3 inches per second (ips) for a maximum input flow of 1.8 gallons per minute (gpm) (see 4.6.3).

3.4.4 Input flow. The maximum required input flow with the servo valve commanded at rated input shall be 1.8 gpm for an actuator piston velocity of 4.3 ips (see 4.6.3).

3.4.5 Rod seal leakage. The total rod seal leakage shall be not greater than four drops per hour under all operating conditions (see 4.6.3).

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NOTES:

1. All dimensions are in inches.
2. Metric equivalents shall be acceptable.

FIGURE 1. Actuator, hydraulic.

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3.4.6 Proof pressure. The actuator shall withstand without evidence of internal or external leakage a proof gauge pressure of 3020 pounds force per square inch (lbf/in^2) simultaneously applied to each chamber (see 4.6.4).

3.4.7 Stalled leakage. The actuator total stalled leakage shall be not greater than 0.05 gpm for a 3000 lbf/in^2 differential pressure across the piston (see 4.6.4).

3.4.8 Stall force. The actuator shall have a stall force not less than 990 pounds with 675 lbf/in^2 differential across the piston and the maximum upstream gauge pressure is 700 lbf/in^2 (see 4.6.5).

3.4.9 Breakaway differential pressure. The maximum differential pressure required to start piston motion in either direction shall be not greater than 14 lbf/in^2 with zero external load (see 4.6.5).

3.4.10 External leakage. There shall be no evidence of external leakage, other than slight wetting insufficient to form a drop (see 4.6.6).

3.5 Environmental service conditions. The actuator shall operate satisfactorily, as specified, during or after exposure to the following environmental conditions (see 4.6.7).

3.5.1 Temperature.

3.5.1.1 Operating. The actuator shall operate as specified while being subjected to a temperature range of -31.7 to 71.1° Celsius (C), (see 4.6.7.1.1).

3.5.1.2 Nonoperating. The actuator shall operate as specified after being subjected to a temperature range of -40 to 71.1°C (see 4.6.7.1.2).

3.5.2 Vibration. The actuator shall operate satisfactorily while being subjected to sinusoidal vibration at an intensity of 0.10 inch double amplitude (da) at a frequency of 5 to 10 Hertz (Hz), ± 0.51 gravity units (g) from 10 to 13 Hz, 0.060 inch da from 13 to 55 Hz and ± 10 g from 55 to 500 Hz (see 4.6.7.2).

3.5.3 Humidity. Humidity tests shall be in accordance with Method 507.3 of MIL-STD-810.

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3.5.4 Sand and dust. Sand and dust tests shall be in accordance with Method 510.3 of MIL-STD-810.

3.5.5 Salt fog. Salt fog tests shall be in accordance with Method 509.3 of MIL-STD-810.

3.5.6 Icing and freezing rain test. These tests shall be in accordance with Method 521.1 of MIL-STD-810 if the actuator is designed such that accumulation of ice on external surfaces will cause malfunction.

3.5.7 Shock. The actuator shall operate satisfactorily after being subjected to three 30 g half sine shocks of 11 milliseconds (ms) duration in each of the three orthogonal axes (see 4.6.7.3).

3.6 Marking. The valve shall be marked in accordance with the requirements of MIL-STD-130 (see 4.6.1).

3.7 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9 (see 4.6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize

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submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Test equipment and inspection facilities. The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the conditions of 4.3.1, 4.3.2, 4.3.3 and 4.3.4.

4.3.1 Temperature. Temperature shall be room ambient, 26.6 ± 4°C.

4.3.2 Barometric pressure. Barometric pressure shall be facility ambient, 28 to 32 inches of mercury (in Hg).

4.3.3 Humidity. The relative humidity shall be not greater than 95 percent.

4.3.4 Fluid temperature and pressure. Fluid temperature shall be 37.8 ± 11.1°C. Supply gauge pressure shall be 1000 + 200, -300 lbf/in^2 with a return gauge pressure of 20 + 5 lbf/in^2.

4.3.5 Inspection equipment. Special inspection equipment for conducting inspections and tests shall consist of a test manifold to simulate ports 1 and 2 of the flow control valve conforming to MIL-V-47308 and a complete flow control valve conforming to MIL-V-47308.

4.4 First article inspection. First article inspection shall consist of the inspection and tests specified in table I. The sequence of inspection shall be at the option of the contractor. Quantity production shall be withheld until the first article sample has been pronounced satisfactory by the procuring activity (see 3.1 and 6.3).

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TABLE I. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Temperature	3.5.1	4.6.7.1
Operating	3.5.1.1	4.6.7.1.1
Nonoperating	3.5.1.2	4.6.7.1.2
Vibration	3.5.2	4.6.7.2
Humidity	3.5.3	4.6.7.4
Sand and dust	3.5.4	4.6.7.5
Salt fog	3.5.5	4.6.7.6
Icing and freezing rain	3.5.6	4.6.7.7

4.5 Quality conformance inspection. Quality conformance inspection shall be conducted on the basis of single sampling of a random sample selected from each inspection lot or less for an acceptable quality level.

4.5.1 Inspection lot. The lot size shall be not greater than the quantity of items produced in one continuous operation utilizing the same production processes for delivery for one contract or order.

4.5.2 Inspection and tests. Samples selected in accordance with 4.5 shall be subjected to the inspections and tests in table II. Sequence of inspections shall be at the option of the contractor.

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TABLE II. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Examination of product	3.3, 3.3.1, 3.3.2, 3.6 and 3.7	4.6.1
Stroke	3.4.1	4.6.2
Displacement polarity	3.4.2	4.6.2
Velocity	3.4.3	4.6.3
Input flow	3.4.4	4.6.3
Rod seal leakage	3.4.5	4.6.3
Proof pressure	3.4.6	4.6.4
Stalled leakage	3.4.7	4.6.4
Stall force	3.4.8	4.6.5
Breakaway differential pressure	3.4.9	4.6.5
External leakage	3.4.10	4.6.6

4.5.2 Sampling.

4.5.2.1 For examination. A random sample shall be selected from each lot in accordance with table III.

4.5.2.2 For acceptance testing. A random sample shall be selected from each lot in accordance with table III.

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TABLE III. Sampling plan for examination and acceptance testing.

Lot Size	Sample Size
2 to 13	All
14 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,001 to 150,000	74
150,001 to 500,000	90
500,001 and over	102
In all cases: The acceptance number is ZERO The rejection number is ONE	

4.6 Methods of inspection.

4.6.1 Examination of product. Each valve shall be visually and dimensionally inspected and weighed to determine conformance to the requirements of this specification (see 3.3, 3.3.1, 3.3.2, 3.6 and 3.7).

4.6.2 Stroke and polarity. With the flow control valve of 4.3.5 installed on the actuator and the pressures of 4.3.4 applied, the valve shall be tested for conformance to 3.4.1 and 3.4.2. The actuator shall be inspected for conformance to 3.4.10.

4.6.3 Velocity, input flow and rod seal leakage. With the flow control valve of 4.3.5 installed on the actuator and the actuator in the test configuration shown on figure 2, the pressures of 4.3.4 shall be applied and the actuator tested for conformance to 3.4.3, 3.4.4 and 3.4.5. The actuator shall be inspected for conformance to 3.4.10.

4.6.4 Proof pressure and stalled leakage. With the test manifold of 4.3.5 installed and the actuator piston locked at midstroke, the actuator shall be proof tested for conformance to 3.4.6. The actuator shall be tested for stalled leakage at midstroke and at positions $\pm$ 0.75 inches from midstroke for

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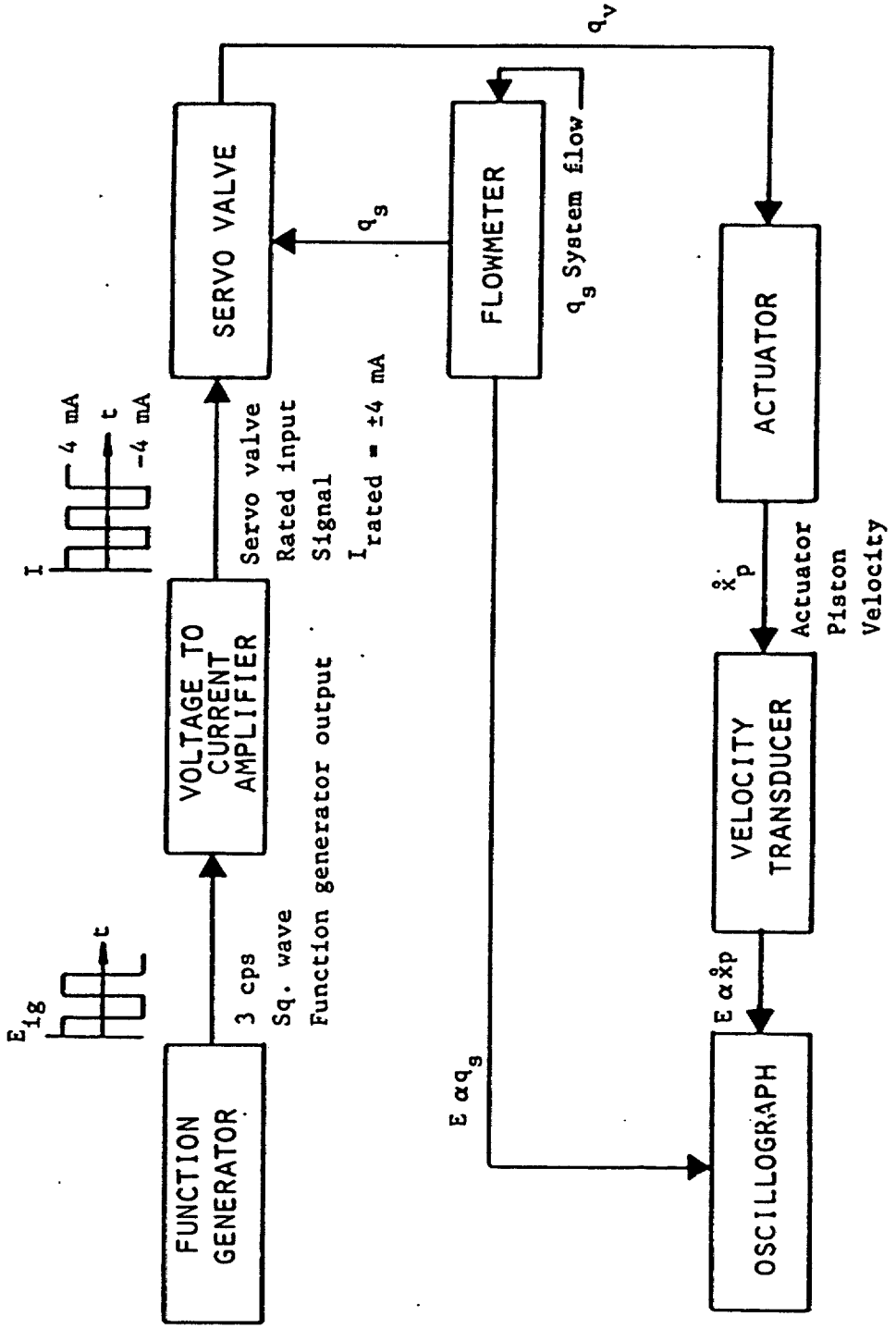


FIGURE 2. Actuator velocity testing, block diagram.

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conformance to 3.4.7. The actuator shall also be inspected for conformance to 3.4.10.

4.6.5 Stalled force and breakaway differential pressure. With the test manifold of 4.3.5 installed and the actuator loaded, the actuator shall be tested extended, 25 percent (%) retracted, 50% retracted, 75% retracted, fully retracted, 25% extended, 50% extended and 75% extended for conformance to 3.4.8. The above will be repeated with the actuator unloaded to determine conformance to 3.4.9. The actuator shall be inspected for conformance to 3.4.10.

4.6.6 External leakage. The actuator shall be inspected for conformance to 3.4.10 after each test.

4.6.7 Environmental service conditions. The actuator shall be tested for conformance to 3.5.

4.6.7.1 Temperature.

4.6.7.1.1 Operating. The actuator shall be tested in accordance with MIL-STD-810, method 501, procedure II to determine conformance to 3.5.1.1 for high temperature. The actuator shall be tested in accordance with MIL-STD-810, method 502, procedure II to determine conformance to 3.5.1.1 for low temperature.

4.6.7.1.2 Nonoperating. The actuator shall be tested in accordance with MIL-STD-810, method 501, procedure I to determine conformance to 3.5.1.2 for high temperature. The actuator shall be tested in accordance with MIL-STD-810, method 502, procedure I to determine conformance to 3.5.1.2 for low temperature.

4.6.7.2 Vibration. The actuator shall be tested in accordance with MIL-STD-810, category 5 to determine conformance to 3.5.2.

4.6.7.3 Shock. The actuator shall be tested in accordance with MIL-STD-810, procedure I to determine conformance to 3.5.7.

4.6.7.4 Humidity. The actuator shall be tested in accordance with 3.5.3.

4.6.7.5 Sand and dust. The actuator shall be tested in accordance with 3.5.4.

4.6.7.6 Salt fog. The actuator shall be tested in accordance with 3.5.5.

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4.6.7.7 Icing and freezing rain. The actuator shall be tested in accordance with 3.5.6.

5. PACKAGING

5.1 Preservation, packing and marking. Preservation, packing and marking shall be level A or B in accordance with MIL-P-45213.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This actuator is intended for use in a military subsystem.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. When first article is required (see 3.1 and 4.4)

d. Conditions for quality conformance inspection (see 4.5)

e. Sampling plan if other than specified (see 4.5.2)

f. Requirements for preservation, packing and marking (see section 5).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the

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Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Control system
Servo valves
Solenoids

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing activity:
Army - MI

(Project No. 1420 - A304)

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-47312A (MI)	2. DOCUMENT DATE (YYMMDD) 921216

3. DOCUMENT TITLE
ACTUATOR, HYDRAULIC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER U.S. ARMY MISSILE COMMAND	b. TELEPHONE (Include Area Code) (1) Commercial: (205) 876-6980 (2) AUTOVON: 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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